

DYNAMIC PROGRAMMING

Masterclass

From Interview Confusion to Crystal Clarity

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What is Dynamic Programming?

The Fundamental Concept

"An algorithmic technique that solves complex problems by breaking them into overlapping sub-problems and storing results to avoid redundant computation."



Overlapping Sub-problems

Same smaller problems solved repeatedly → store results



Optimal Substructure

Optimal solution built from optimal sub-solutions



Memoization / Tabulation

Top-down (cache) or Bottom-up (table) storage

Your DP Learning Roadmap

What You Should Be Studying

01

Recursion

Master base cases,
recursive tree thinking

02

Memoization

Top-down DP with cache /
HashMap

03

Tabulation

Bottom-up DP with 1D / 2D
arrays

04

Space Opt.

Reduce $O(n^2)$ space to $O(n)$
or $O(1)$

05

Patterns

Knapsack, LCS, LIS, MCM,
Grid, Trees

 *Pro Tip: Practice each level with at least 10 problems before moving to the next step.*

Core DP Concepts

Build Your Mental Model

State Definition

What does $dp[i]$ represent? The single most important step in any DP solution.

Recurrence Relation

How do we compute $dp[i]$ from smaller states? Express it mathematically.

Base Cases

Smallest sub-problems with known answers. Guards against out-of-bound access.

State Space & Dimensions

1D, 2D, or 3D table. Each dimension captures a variable in your problem.

Transition Order

Top-down uses recursion. Bottom-up needs careful loop ordering.

Space Optimization

Often only the last row/column is needed. Reduce to $O(n)$ with rolling array.

Must-Know DP Patterns for Interviews

Identify the pattern → Apply the template → Win

0/1 Knapsack

Subset Sum, Partition Equal Subset, Target Sum

Unbounded Knapsack

Coin Change, Rod Cutting, Ribbon Cutting

Fibonacci Style

Climbing Stairs, House Robber, Jump Game

Longest Common Subseq

LCS, Edit Distance, Shortest Common Superseq

Longest Incr. Subseq

LIS, Max Envelopes, Russian Doll Envelopes

Matrix Chain / MCM

Burst Balloons, Palindrome Partition, Egg Drop

Grid / Path DP

Unique Paths, Dungeon Game, Cherry Pickup

Interval DP

Burst Balloons, Strange Printer, Zuma Game

Top 15 DP Problems for Interviews

Every Serious Candidate Must Solve These

#	Problem	Pattern	Difficulty
LC 70	Climbing Stairs	Fibonacci	Easy
LC 198	House Robber	Fibonacci	Medium
LC 322	Coin Change	Unbounded KS	Medium
LC 300	Longest Incr. Subsequence	LIS	Medium
LC 1143	Longest Common Subsequence	LCS	Medium
LC 72	Edit Distance	LCS	Medium
LC 416	Partition Equal Subset Sum	0/1 Knapsack	Medium
LC 62	Unique Paths	Grid DP	Medium
LC 64	Minimum Path Sum	Grid DP	Medium
LC 312	Burst Balloons	Interval DP	Hard
LC 115	Distinct Subsequences	LCS variant	Hard
LC 97	Interleaving String	2D DP	Medium
LC 10	Regular Expression Match	2D DP	Hard
LC 85	Maximal Rectangle	Stack + DP	Hard
LC 514	Freedom Trail	State Machine	Hard

Memoization vs Tabulation

Two Sides of the Same DP Coin – Fibonacci Example

TOP-DOWN (Memoization)

```
# Python - Top-Down

memo = {}

def fib(n):

    if n <= 1:

        return n

    if n in memo:

        return memo[n]  # ← cache hit

    memo[n] = fib(n-1) + fib(n-2)

    return memo[n]

# Time: O(n)   Space: O(n)
```

BOTTOM-UP (Tabulation)

```
# Python - Bottom-Up

def fib(n):

    if n <= 1:

        return n

    dp = [0] * (n + 1)  # table

    dp[1] = 1

    for i in range(2, n+1):

        dp[i] = dp[i-1] + dp[i-2]

    return dp[n]

# Space-Opt → O(1) space:

a, b = 0, 1

for _ in range(2, n+1): a,b=b,a+b
```

Frequently Asked Interview Questions

What FAANG & Top Companies Actually Ask

? How do you identify if a problem requires DP?

Look for: optimal value (min/max), counting solutions, overlapping sub-problems, choices at each step.

? What's the difference between DP and Divide & Conquer?

D&C sub-problems are independent; DP sub-problems overlap. DP stores answers to avoid recomputation.

? When to use top-down vs bottom-up DP?

Top-down is intuitive & skips unused states. Bottom-up avoids recursion stack overhead & is cache-friendly.

? How do you optimize space in DP?

Observe which previous rows/columns are needed. Use rolling arrays or two variables for 1D Fibonacci-style.

? Explain the Knapsack problem and its variants.

0/1: each item used once. Unbounded: unlimited use. Fractional: greedy. Bounded: limited count per item.

? What is the time complexity of LCS / Edit Distance?

$O(m \times n)$ time and $O(m \times n)$ space naively; reducible to $O(\min(m,n))$ space with rolling array technique.

Dr. Amar's 5-Step DP Framework

Apply this to EVERY DP problem in your interview

1

Define the State

$dp[i]$ = ? What does this index represent? Write it down.

2

Identify Choices

At each state, what decisions can we make? Include / Exclude.

3

Write Recurrence

Express $dp[i]$ in terms of previous states. Be explicit.

4

Handle Base Cases

Smallest inputs with known values. Prevents infinite recursion.

5

Compute & Return

Fill table bottom-up or recurse top-down. Return $dp[\text{target}]$.

30-Day DP Interview Prep Plan

Structured Practice – Career Credentials Recommended

Week 1

Foundations

► Recursion: Factorial, Fibonacci, Power Set

► Memoization: Top-down Fibonacci, Climbing Stairs

► Tabulation: House Robber, Min Cost Climbing Stairs

► Time/Space Complexity Analysis for each

Week 2

1D DP Patterns

► Fibonacci Pattern: Decode Ways, Jump Game

► Unbounded KS: Coin Change, Coin Change II

► Partition: Subset Sum, Equal Partition

► LIS: Longest Incr. Subsequence, Max Envelopes

Week 3

2D DP Patterns

► LCS: Edit Distance, Distinct Subsequences

► Grid DP: Unique Paths, Min Path Sum

► Palindrome DP: Longest Palindromic Subseq.

► String DP: Interleaving String, Wildcard Match

Week 4

Hard & Mock Interviews

► Interval DP: Burst Balloons, Strange Printer

► Tree DP: House Robber III, Binary Tree Cameras

► State Machine: Best Time to Buy/Sell Stock series

► 5 full timed mock interview sessions

You're Now Ready to

Master Dynamic Programming

- ✓ Understand the 2 core properties of DP
- ✓ Know all 8 key patterns for interviews
- ✓ Apply the 5-step framework on any problem
- ✓ Have a 30-day structured practice plan



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